Review plan · for the teacher

Circuits and microcontrollers: unit check

IoT · Level 2 · Unit check

Time	Materials
45 min	Board, chalk, paper. No devices needed.

Learning goals

- Current, LEDs and resistors
- Logic: switches that decide
- Bits and bytes
- Microcontrollers: sense, decide, act

Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Write the hardest question on the board for tomorrow

Discussion

- Which idea will you use first?
- Which question surprised you, and why?
- What will your pupils build or measure first?

Exit ticket

- One idea I will use, and when I will use it

Try it at school tomorrow
Pick the idea from this unit your pupils need most, teach it this week, and tell a colleague how it went.

Quiz sheet · for each team

Circuits and microcontrollers: unit check

IoT · Level 2 · Unit check

Team _____ Date _____

1 **A torch with new batteries won't light. What should you check first?**

◯ Whether a bigger battery would push more current ◯ The loop: wires, contacts and battery direction
◯ Whether the bulb needs a resistor to protect it

2 **Match each wiring or device to its gate.**

Two switches in series	• • • •	NOT
Two switches in parallel	• • • •	AND
A fridge-door light switch	• • • •	XOR
A stair light with a switch at each end	• • • •	OR

3 **Choose the resistor for a red LED on two AA batteries.**

3 V from the batteries minus 2 V for the LED leaves _____ V. For 10 mA, the resistor should be _____ Ω.

Word bank: 3 · 1 · 300 · 100

4 **A garden pump runs when "soil dry AND NOT raining". The soil is dry and it is raining, so the pump runs.**

◯ True ◯ False

5 **Which statements about a microcontroller are true?**

☐ An analog pin turns a voltage into a number ☐ Setup runs once, at power-on
☐ The loop runs again and again ☐ A pin can power a pump directly ☐ It needs the internet to work

6 **Put these in order of size, smallest first. You'll need to work one of them out.**

☐ One full SMS
☐ A phone photo
☐ One Arabic letter in UTF-8
☐ A day of readings, one every 10 minutes, 2 bytes each
☐ A page of a book

7 **Which binary number is 9?**

◯ 1100 ◯ 0110 ◯ 1010 ◯ 1001

8 **An Arabic SMS holds fewer letters than an English one because each Arabic letter takes more bytes.**

◯ True ◯ False

9 **Your school has no electronics kit. Which are good ways to teach this unit?**

☐ Human logic gates with 1 and 0 cards ☐ Skipping the practical part ☐ Binary dot cards
☐ A lamp from home on the mains, to show a real circuit ☐ A pupil playing the microcontroller
☐ A free simulator on a phone

10 **A data logger reads a light sensor with 10 bits. A new board uses 12 bits. What changes?**

◯ The same range is split into finer steps ◯ Readings are sent faster, with more bits each
◯ It can measure brighter light than before

Answers · for the teacher

Circuits and microcontrollers: unit check

IoT · Level 2 · Unit check

1 **The loop: wires, contacts and battery direction**

Current flows only round a closed loop, so a loose wire, a dirty contact or a battery in backwards keeps it dark.

2 **Two switches in series — AND**
Two switches in parallel — OR
A fridge-door light switch — NOT
A stair light with a switch at each end — XOR

In series, both switches must close (AND); in parallel, either will do (OR).

3 **3 V from the batteries minus 2 V for the LED leaves 1 V. For 10 mA, the resistor should be 100 Ω.**

The LED takes about 2 V, so 1 V is left for the resistor; R = V ÷ I = 1 ÷ 0.01 = 100 Ω.

4 **False**

Raining is 1, so NOT raining is 0, and 1 AND 0 is 0: the pump stays off.

5 **Setup runs once, at power-on; The loop runs again and again; An analog pin turns a voltage into a number**

Setup, the loop and analog inputs are how every microcontroller works.

6 **Top to bottom: 2, 5, 1, 3, 4**

2 bytes; 140 bytes; 144 × 2 = 288 bytes; about 2,000 bytes; about 3,000,000 bytes.

7 **1001**

9 = 8 + 1, so the 8s place and the 1s place are on: 1001.

8 **True**

An SMS carries 140 bytes. Latin letters are packed into 7 bits each (160 of them); Arabic letters take 2 bytes each, so only 70 fit.

9 **Human logic gates with 1 and 0 cards; Binary dot cards; A pupil playing the microcontroller; A free simulator on a phone**

Every idea in this unit can be acted out, drawn or simulated.

10 **The same range is split into finer steps**

Two more bits mean four times as many levels (0 to 4095 instead of 0 to 1023) across the same range, so each step is smaller.